Sensor Selection for Embedded Systems

From physical world to digital values

IMU • Environmental • Optical • Current • Hall

Audience: Beginner → Intermediate Engineering Students

Pick the right sensor • Interface • Calibrate • Filter

Why Sensor Selection Matters

The sensor is your contact with reality

Resolution

Can you measure the change you care about?

Accuracy

How close to truth at all conditions?

Drift

Does it stay accurate over time & temperature?

Interface

I²C, SPI, analog, PWM — MCU pin budget?

Power

Always-on µA budget for battery devices.

Cost

BOM hit — passive or sensor IC?

Sensor Categories

What students will encounter in industry

Motion / IMU

Acceleration, rotation,
magnetic heading.

Environmental

Temperature, humidity,
pressure, gas.

Optical / Imaging

Ambient light, proximity,
image sensor.

Position / Distance

Encoders, ToF, ultrasonic,
LIDAR.

Current / Voltage

Power monitoring,
battery gauge.

Bio / Special

ECG, EMG, pulse, oximeter,
strain.

IMU — Inertial Measurement Unit

Accelerometer + gyro + (sometimes) magnetometer

IC	DOF	Interface	Notes
MPU-6050	6	I^2C	Cheap & common; legacy
MPU-9250	9	I^2C/SPI	6-axis + magnetometer (EOL)
ICM-42688	6	I^2C/SPI	Modern, low-noise
BMI270	6	I^2C/SPI	Bosch — wearable / phone
BNO055	9	I^2C	Built-in sensor fusion
LSM6DSOX	6	I^2C/SPI	ST — programmable FSM

Tip

BNO055 has hardware sensor fusion (Bosch's BSX) — gives roll/pitch/yaw without you writing the maths. Great for drone/robot students.

Environmental Sensors

Temperature, humidity, pressure, gas

IC	Measures	Interface	Use case
BME280	T / H / P	I^2C/SPI	Weather, IoT — most popular
BME680	T / H / P / VOC	I^2C/SPI	+ Air quality
SHT41	T / H	I^2C	Higher accuracy than BME
SCD41	CO_2 + T + H	I^2C	True NDIR CO_2 sensor
DS18B20	T only	1-Wire	Cheap, robust, long cable
LM75 / TMP102	T only	I^2C	Cheap IC thermometer

Optical & Proximity Sensors

Light, distance, imaging

Ambient light

BH1750, OPT3001 —
0-100 klx range.

Proximity

VCNL4040 — IR distance
< 200 mm.

ToF distance

VL53L0X / L1X / L3CX —
10-2000 mm.

Heart-rate / SpO_2

MAX30102 — pulse +
blood-oxygen.

Colour

TCS3472 — RGB + clear
for recognition.

Image

OV2640 / OV5640 —
CMOS camera modules.

Position & Encoders

Knowing where you are

- Incremental rotary encoders — A/B phase output, count edges in MCU timer
- Absolute rotary encoders — gives angle directly, SPI/I^2C readout
- Magnetic rotary IC — AS5600 (I^2C), AS5048A (SPI/PWM)
- Linear quadrature encoders — same A/B principle on a track
- Resolver — automotive grade, very robust to dirt and heat
- Hall switches — simple on/off, position sensing on motors
- Ultrasonic (HC-SR04) — cheap distance, 2 cm – 4 m
- ToF / LIDAR — sub-mm to 10 m, for AR/robots

Current Sensing

How much current is the load drawing?

Shunt + INA

INA219 / INA240 / INA228 —
I²C readout.

Hall current IC

ACS712 / TMCS1100 —
isolated, ± current.

CT (clamp)

Current transformer for
AC mains (50/60 Hz).

Rogowski coil

Flexible AC sensor for
high-current pulses.

Op-amp + shunt

DIY low-side; cheapest,
needs calibration.

PMIC built-in

Many battery PMICs
have coulomb counter.

Sensor Interfaces

How to talk to a sensor

Interface	Pins	Speed	Best for
I²C	2 (SDA/SCL)	100k-3.4M	Most ICs, multi-drop
SPI	4+ (MOSI/MISO/SCK/CS)	up to 50 MHz	Fast, single sensor
Analog	1 (V_OUT)	ADC limited	Cheap sensors, thermocouples
PWM / Pulse	1	kHz typical	HC-SR04, simple encoders
UART	2 (TX/RX)	115k typical	GPS, modem-style
1-Wire	1	16k	DS18B20 long cable
CAN	2 (CANH/CANL)	1 Mbit	Automotive sensor buses

Calibration

Raw values rarely tell the truth

- Zero-offset calibration: measure with no input, subtract bias
- Two-point calibration: known low + known high → linear interpolation
- Temperature compensation: many sensors drift; correct in firmware
- Magnetometer hard/soft-iron calibration: rotate device, fit ellipsoid
- IMU: stationary calibration to zero out gyro bias
- Sensor fusion: combine accelerometer + gyro to cancel drift (Mahony, Madgwick, Kalman)
- Store calibration constants in flash / EEPROM — re-apply at boot

Filtering & Sampling

Get clean numbers from noisy sensors

Moving average

Average last N samples.
Simple, smooths spikes.

Low-pass IIR

$y[n] = \alpha x[n] + (1-\alpha) y[n-1]$.
Cheap, single state.

FIR filter

Linear-phase, programmable.
More CPU cost.

Kalman

Optimal under Gaussian noise.
Requires modeling.

Complementary

HF gyro + LF accelerometer.
Classic IMU fusion.

Median

Resistant to outliers.
Great for ToF.

PCB & Mechanical Mounting

Most sensor errors come from layout & mounting

- IMU: rigidly attached to chassis, axes aligned to body, no flex
- Temperature: thermally isolated from MCU and heaters; cut-outs help
- Humidity / VOC: airflow through enclosure
- Pressure / barometer: small port hole, gore-tex membrane to keep water out
- Optical sensors: lens / window glass tested for transmission
- Hall current: 50 % overlap of magnet field — vendor recommends position
- Routing: keep digital clocks far from analog sensor traces
- Shielding: cans over noise-sensitive sensors near switching circuits

Common Sensor Mistakes

Caught only after the prototype lab gives weird numbers

- Forgetting pull-up resistors on I²C SDA/SCL
- Using a single-ended ADC for a differential sensor
- Ignoring ADC reference noise (VREF = VCC is usually bad)
- Sampling slower than the sensor's update rate (aliased data)
- Not handling sensor warm-up time — first 100 ms readings are garbage
- No calibration storage — every reboot resets bias
- Mounting IMU on a flexing PCB section (rigid-flex bend area)
- Sharing AVDD with motor driver / RF — noisy readings

Selection Workflow

How to pick a sensor in 30 minutes

- 1. Define what you measure & target precision
- 2. Search DigiKey / Mouser parametric for that category + spec
- 3. Compare top 3-5 candidates by accuracy, power, interface, cost
- 4. Check breakout boards on Adafruit / SparkFun — quick prototype
- 5. Read datasheet thoroughly — operating conditions, calibration notes
- 6. Verify supply ecosystem (Indian distributor in stock?)
- 7. Order eval — measure your actual scenario, not lab conditions
- 8. Plan calibration & firmware filtering before designing PCB

Key Takeaways

Key takeaways from this lecture

Spec your need first

Resolution + accuracy + drift.

Interface matters

I²C vs SPI vs analog drives MCU choice.

Calibrate everything

Raw values rarely meet datasheet specs.

Filter in firmware

Moving avg, IIR, fusion as needed.

Mount carefully

Most errors come from physical setup.

Use breakout boards

Validate before designing PCB.

Recommended Resources

Sensor learning paths

- Adafruit / SparkFun / Pimoroni — breakout boards + tutorials
- Bosch / ST / TI / Sensirion sensor datasheets & app notes
- Edge Impulse — ML on sensor data
- Robert Mahony / Sebastian Madgwick — IMU fusion papers (free)
- EEVblog forum — practical sensor debugging threads
- Digi-Key / Mouser parametric search for filtering candidates
- AnalogIn group on Reddit — analog frontend discussion
- Sensirion / Bosch annual application reports

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- Adafruit / SparkFun tutorial libraries
- Bosch / ST / TI / Sensirion datasheets
- Edge Impulse — TinyML for sensor data
- Madgwick / Mahony fusion papers
- Kalman filter intro — Tim Babb's tutorial
- Sensor Fusion course (Coursera, by Lerstad)
- EEVblog — sensor measurement threads
- Industry contacts: Bosch India, ST India, Sensirion India

Thank you • Build something this week.